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The Macroscopic Coherent State

3.1 Introduction

We have seen in the previous chapters that the concept of the macroscopic wave function, $\psi(\mathbf{r})$, is central to understanding atomic Bose-Einstein condensates, superfluid ^4He , and even superconductivity within the Ginzburg-Landau theory. But the connection between these ideas is not at all clear, since the atom condensates and ^4He are bosonic systems, while superconductivity is associated with the conduction electrons in metals which are fermions. The physical meaning of the Ginzburg-Landau order parameter was not at all clear until after 1957 when Bardeen Cooper and Schrieffer (BCS) published the first truly microscopic theory of superconductivity. Soon afterwards the connection was finally established by Gorkov. He was able to show that, at least in the range of temperatures near T_c , the Ginzburg Landau theory can indeed be derived from the BCS theory. Furthermore this provides a physical interpretation of the nature of the order parameter. Essentially it is describing a macroscopic wave function, or condensate, of Cooper pairs.

The purpose of this chapter is to clarify the concept of a macroscopic wave function, and show how it arises naturally from the physics of **coherent states**. Coherent states were first developed in the field of quantum optics, and were especially useful in the theory of the laser. The laser is, of course, yet another type of macroscopic coherent state, with close similarities to atomic BEC. In this chapter we shall first introduce the concept and mathematical properties of coherent states before applying them to bosonic systems. Using this approach we shall rederive the Gross-Pitaevskii equations for the weakly interacting Bose gas, as originally introduced above in Chapter 1.

Coherent states can be defined for fermions as well as for bosons. But single fermion coherent states, while very useful in other contexts, are not directly useful in the theory of superconductivity. What is needed is a coherent state of **fermion pairs**. Such coherent states are exactly the type of many-body quantum state first written down in the theory of Bardeen Cooper and Schrieffer in their 1957 theory of superconductivity. We shall

postpone a full discussion of this BCS theory until the following chapter. Here we focus specifically on the physics of the BCS coherent state, without worrying, for example, about why it is a stable ground state. This separation has the advantage that the key concepts can be presented more clearly and they can be seen to be very general. Indeed it is not necessary to rely on every detail of the BCS theory in order to understand physically the properties of the quantum coherent state. In this approach we can also see very generally the connection between the BCS state and the Ginzburg-Landau theory, since the coherent state of electron pairs provides a direct connection to the order parameter $\psi(\mathbf{r})$.

This logical separation between the full detail of the BCS theory and the physical origin of the order parameter is not just an educational device; it also has a useful purpose more generally. For example, there are several superconductors where we do not know if the BCS theory is applicable, the high T_c superconductors being the most obvious example. However, even though we do not know the **mechanism of pairing** we do know that these systems do have Cooper pairs. For example they have flux quantization in the usual units of $\Phi_0 = h/2e$, showing that the fundamental charge unit is $2e$. We can also assume that there will be a Ginzburg-Landau order parameter whatever the actual pairing mechanism, and this knowledge will provide a sound basis for many theories of the superconducting state (eg theories of the vortex matter states in high T_c superconductors). We can therefore separating the actual mechanism of pairing from its main consequence: the existence of the order parameter.

3.2 Coherent states

To start with, let us just go back to some elementary undergraduate quantum mechanics: the quantum Harmonic oscillator. The Hamiltonian operator is

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega_c^2}{2}x^2 \quad (3.1)$$

where $\hat{p} = i\hbar \frac{d}{dx}$ is the one-dimensional momentum operator, m is the particle mass and ω_c is the classical oscillator angular frequency. The eigenstates and of the oscillator are given by

$$\hat{H}\psi_n(x) = E_n\psi_n(x), \quad (3.2)$$

with energy levels E_n .

The most elegant method of solving this classic problem, to find E_n and $\psi_n(x)$ is to introduce the **ladder operators**,

$$\hat{a}^+ = \frac{1}{(\hbar\omega_c)^{1/2}} \left(\frac{\hat{p}}{(2m)^{1/2}} - i \frac{(m\omega_c^2)^{1/2}x}{(2)^{1/2}} \right)$$

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$$\hat{a}^+ = \frac{1}{(\hbar\omega_c)^{1/2}} \left(\frac{\hat{p}}{(2m)^{1/2}} + i \frac{(m\omega_c^2)^{1/2}x}{(2)^{1/2}} \right). \quad (3.3)$$

These operators have a number of very useful and easily derived properties,¹ which can be summarized as follows:

$$\hat{a}^+ \psi_n(x) = (n+1)^{1/2} \psi_{n+1}(x) \quad (3.4)$$

$$\hat{a} \psi_n(x) = n^{1/2} \psi_{n-1}(x) \quad (3.5)$$

$$\hat{a}^+ \hat{a} \psi_n(x) = n \psi_n(x) \quad (3.6)$$

$$[\hat{a}, \hat{a}^+] = 1. \quad (3.7)$$

The first of these relations shows that the operator $\hat{a}^+$ changes any state to the next one higher up the “ladder” of the possible n values. Similarly second shows that $\hat{a}$ moves down the ladder. From the third relation, the combination $\hat{a}^+ \hat{a}$ results in no change of n . Therefore Eq. 3.6 shows that we can identify the combination $\hat{n} = \hat{a}^+ \hat{a}$ as the **number operator**, which gives the quantum number n of any state,

$$\hat{n} \psi_n(x) = n \psi_n(x). \quad (3.8)$$

The commutator relation, Eq. 3.7,

$$[\hat{a}, \hat{a}^+] = \hat{a} \hat{a}^+ - \hat{a}^+ \hat{a} = 1 \quad (3.9)$$

is fundamental to the quantum mechanics of bosonic systems, as we shall see below.

In terms of the latter operators the oscillator Hamiltonian is given by,

$$\hat{H} = \hbar\omega_c \left(\hat{a}^+ \hat{a} + \frac{1}{2} \right). \quad (3.10)$$

Combined with Eq. 3.6 it immediately shows that the energy levels are

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right). \quad (3.11)$$

exactly as expected.

Using the ladder raising operator, $\hat{a}^+$ repeatedly, Eq. shows that one can construct all of the eigenvectors, $\psi_n(x)$, iteratively by acting repeatedly on an initial ground state $\psi_0(x)$,

$$\psi_n(x) = \frac{1}{(n!)^{1/2}} (\hat{a}^+)^n \psi_0(x). \quad (3.12)$$

Therefore to find the complete set of states it is only necessary to find $\psi_0(x)$ (which elementary quantum mechanics tells us is a simple gaussian

¹See exercise 5.1.

function), and then all of the remaining quantum states can be generated essentially automatically.

But the ladder operators also have many other uses. In particular let us define a **coherent state** by,

$$|\alpha\rangle = C \left(\psi_0(x) + \frac{\alpha}{1!^{1/2}} \psi_1(x) + \frac{\alpha^2}{2!^{1/2}} \psi_2(x) + \frac{\alpha^3}{3!^{1/2}} \psi_3(x) + \dots \right) \quad (3.13)$$

where α is any arbitrary complex number, and C is a normalization constant. This constant C can be found easily from the normalization condition

$$\begin{aligned} 1 &= \langle \alpha | \alpha \rangle \\ &= |C|^2 \left(1 + \frac{|\alpha|^2}{1!} + \frac{(|\alpha|^2)^2}{2!} + \frac{(|\alpha|^2)^3}{3!} + \dots \right) \\ &= |C|^2 e^{|\alpha|^2} \end{aligned} \quad (3.14)$$

and so we can take $C = e^{-|\alpha|^2/2}$.

Coherent states have many interesting properties. The following expression is a particularly useful relation

$$|\alpha\rangle = e^{-|\alpha|^2/2} \left(1 + \frac{\alpha \hat{a}^+}{1!} + \frac{(\alpha \hat{a}^+)^2}{2!} + \frac{(\alpha \hat{a}^+)^3}{3!} + \dots \right) |0\rangle \quad (3.15)$$

where $|0\rangle = \psi_0(x)$ is the ground state and is also the coherent state with $\alpha = 0$. This expression can be written very compactly as,

$$|\alpha\rangle = e^{-|\alpha|^2/2} \exp(\alpha \hat{a}^+) |0\rangle. \quad (3.16)$$

Note that the exponential of any operator, $\hat{X}$ is just defined by the usual series expansion of exponential

$$\exp(\hat{X}) = 1 + \frac{\hat{X}}{1} + \frac{\hat{X}^2}{2!} + \frac{\hat{X}^3}{3!} + \dots \quad (3.17)$$

Another interesting relation which can be obtained from Eq. 3.13 is

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \quad (3.18)$$

therefore they are eigenstates of the ladder operator $\hat{a}$. To prove this we can write the state $\hat{a}|\alpha\rangle$ explicitly

$$\hat{a}|\alpha\rangle = e^{-|\alpha|^2/2} \hat{a} \left(\psi_0(x) + \frac{\alpha}{1!^{1/2}} \psi_1(x) + \frac{\alpha^2}{2!^{1/2}} \psi_2(x) + \frac{\alpha^3}{3!^{1/2}} \psi_3(x) + \dots \right). \quad (3.19)$$

But $\hat{a}\psi_n(x) = n^{1/2}\psi_{n-1}(x)$ and so this gives

$$\hat{a}|\alpha\rangle = e^{-|\alpha|^2/2} \left(0 + \frac{\alpha 1^{1/2}}{1!^{1/2}} \psi_0(x) + \frac{\alpha^2 2^{1/2}}{2!^{1/2}} \psi_1(x) + \frac{\alpha^3 3^{1/2}}{3!^{1/2}} \psi_2(x) + \dots \right) \quad (3.20)$$

which is clearly equal to $\alpha|\alpha\rangle$.

Finally, two further nice properties are also simple consequences of Eq. 3.18,

$$\langle \alpha | \hat{a} | \alpha \rangle = \alpha \quad (3.21)$$

$$\langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = |\alpha|^2. \quad (3.22)$$

Therefore the value of $|\alpha|^2$ gives the mean number operator $\langle \hat{n} \rangle$ of the quantum state. Extending this to $\hat{n}^2$ we can find the number uncertainty Δn ,

$$\begin{aligned} \langle \hat{n}^2 \rangle &= \langle \alpha | \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} | \alpha \rangle \\ &= \langle \alpha | \hat{a}^\dagger (\hat{a}^\dagger \hat{a} + 1) \hat{a} | \alpha \rangle \\ &= |\alpha|^4 + |\alpha|^2 \end{aligned} \quad (3.23)$$

$$\begin{aligned} \Delta n &= \sqrt{\langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2} \\ &= |\alpha|. \end{aligned} \quad (3.24)$$

Coherent states do not have a definite value of the quantum number n , simply because they are not eigenstates of the number operator. In fact the probability of observing the value of n in a quantum measurement of state $|\alpha\rangle$ is actually a Poisson distribution shown in Fig. 3.1

$$P_n = \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2} \quad (3.25)$$

as can easily be seen from Eq. 3.13. In this distribution, as given by Eq. 3.24, the standard deviation in the number n is

$$\Delta n = \sqrt{\langle \hat{n} \rangle} \quad (3.26)$$

or

$$\frac{\Delta n}{\langle n \rangle} \sim \frac{1}{\sqrt{\langle n \rangle}}. \quad (3.27)$$

We shall be mostly interested in **macroscopic** coherent states, where $\langle n \rangle$ is essentially infinite. For such states, one can see that the standard deviation Δn becomes essentially negligible compared to $\langle n \rangle$. Therefore to a good approximation we can approximate many operator expectation values by their **mean-field** values derived from the replacement $\hat{n} \approx \langle n \rangle$. In Fig. 3.1 one can see that the distribution becomes strongly peaked about its mean value of $\langle n \rangle$ even for quite small values of $\langle n \rangle$.

Importantly, the coherent state $|\alpha\rangle$ does have a definite **phase**, θ , though not have a definite quantum number n . The coherent state can be defined for any complex number α ,

$$\alpha = |\alpha| e^{i\theta}. \quad (3.28)$$

Rewriting Eq. 3.13 in terms of these variables we see that

$$|\alpha\rangle = C \left(\psi_0(x) + e^{i\theta} \frac{|\alpha|}{1!^{1/2}} \psi_1(x) + e^{2i\theta} \frac{|\alpha|^2}{2!^{1/2}} \psi_2(x) + \dots \right). \quad (3.29)$$

We see that the term containing ψ_n depends on $e^{in\theta}$. Differentiating with respect to θ one can see that

$$\frac{1}{i} \frac{\partial}{\partial \theta} |\alpha\rangle = \hat{n} |\alpha\rangle. \quad (3.30)$$

But since the states $|\alpha\rangle$ are a complete set (actually an overcomplete set!), we can make the operator identification

$$\frac{1}{i} \frac{\partial}{\partial \theta} = \hat{n}. \quad (3.31)$$

So the phase θ and the number n are **conjugate operators**, in a similar way to momentum and position in standard quantum mechanics. It is possible to state a form of the uncertainty principle for these operators²

$$\Delta n \Delta \theta \geq \frac{1}{2}. \quad (3.32)$$

Coherent states have a fixed phase, but do not have definite values of number n . In contrast the energy eigenstates, $\psi_n(x)$, have a well defined value of n , but have an arbitrary (or ill-defined) phase.

Coherent states have many other beautiful mathematical properties, which we will not have time to explore in detail. In particular they are an overcomplete set, since they can be defined for any point in the complex α plane, and so there are uncountably infinite many such states. They are also not orthogonal, and it is easy to show that (exercise 5.2)

$$|\langle \alpha | \beta \rangle|^2 = e^{-|\alpha - \beta|^2}. \quad (3.33)$$

We can interpret this in terms of the Argand diagram for the complex number α , shown in Fig. 3.2. Every point in the plane represents a valid coherent state. Coherent states at neighbouring points are not orthogonal, but their overlap dies off when $|\alpha - \beta| \sim 1$. Therefore there is effectively one “independent” orthogonal quantum state per unit area of the complex plane. Using polar coordinates, $|\alpha|, \theta$ we see that an element of area contains exactly one quantum state if

$$1 \sim |\alpha| \Delta \theta \Delta |\alpha| \quad (3.34)$$

which is of order $2\Delta n \Delta \theta$, since $\langle n \rangle = |\alpha|^2$. Therefore the number-phase uncertainty principle gives essentially the minimum area per quantum state in the complex α plane of Fig. 3.2.

²A precise proof of this uncertainty relation is a little more tricky than the usual Heisenberg momentum-position uncertainty principle, since θ is only strictly defined in the range between 0 and 2π .

3.3 Coherent States and the Laser

Coherent were first studied extensively in the theory of the laser. In quantum optics the use of the operators $a_{\mathbf{k}s}^+$ and $a_{\mathbf{k}s}$ correspond directly to **creation** and **annihilation** of photons in a particular mode of electromagnetic radiation with wave number $\mathbf{k}$ and polarization s (i.e. left or right circularly polarized). A general quantum state of the system can be represented in the **occupation number representation**

$$|n_{\mathbf{k}_0 s_0}, n_{\mathbf{k}_1 s_1}, n_{\mathbf{k}_2 s_2}, n_{\mathbf{k}_3 s_3}, \dots\rangle \quad (3.35)$$

where $\mathbf{k}_0, \mathbf{k}_1, \mathbf{k}_2$ etc. count all the different plane wave states of the system, eg the cavity of the laser.

In the case of light, the creation and annihilation operators arise quite naturally when one quantizes the electromagnetic radiation field. Each specific mode of the classical radiation field $\mathbf{k}, s$ obeys Maxwell's equations. When these equations are quantized each mode becomes an independent quantum harmonic oscillator. The quantum states, $n_{\mathbf{k}s}$, of each oscillator are interpreted as the number of "photons" present. The creation operator adds a photon while the annihilation operator destroys one,³

$$a_{\mathbf{k}s}^+ |\dots n_{\mathbf{k}s} \dots\rangle = (n_{\mathbf{k}s} + 1)^{1/2} |\dots n_{\mathbf{k}s} + 1 \dots\rangle \quad (3.36)$$

$$a_{\mathbf{k}s} |\dots n_{\mathbf{k}s} \dots\rangle = (n_{\mathbf{k}s})^{1/2} |\dots n_{\mathbf{k}s} - 1 \dots\rangle, \quad (3.37)$$

in complete analogy with the harmonic oscillator ladder operators. From these one can deduce that the same commutation law as the ladder operators must apply. But the operators for independent radiation field modes must commute, and therefore we can write,

$$[a_{\mathbf{k}s}, a_{\mathbf{k}'s'}^+] = \delta_{\mathbf{k}s, \mathbf{k}'s'} \quad (3.38)$$

$$[a_{\mathbf{k}s}, a_{\mathbf{k}'s'}] = 0 \quad (3.39)$$

$$[a_{\mathbf{k}s}^+, a_{\mathbf{k}'s'}^+] = 0. \quad (3.40)$$

In the case of the laser we can naturally shift from the occupation number representation to a coherent state representation. A general coherent state is of the form

$$|\alpha_{\mathbf{k}_0 s_0}, \alpha_{\mathbf{k}_1 s_1}, \alpha_{\mathbf{k}_2 s_2}, \alpha_{\mathbf{k}_3 s_3}, \dots\rangle \equiv e^{-\sum |\alpha_{\mathbf{k}s}|^2/2} \exp\left(\sum_{\mathbf{k}s} \alpha_{\mathbf{k}s} a_{\mathbf{k}s}^+\right) |0\rangle. \quad (3.41)$$

Here $|0\rangle$ is the vacuum state, with no photons present. An ideal coherent laser source is one in which just one of these modes has a macroscopic occupation, $\langle \hat{n}_{\mathbf{k}s} \rangle = |\alpha_{\mathbf{k}s}|^2$ while the others have essentially zero occupation.

³For convenience we will no longer write these as $\hat{a}^+$ and $\hat{a}$, but just as a^+ and a . No ambiguity will arise from this simplified notation, but one must not forget that these are operators and do not commute.

In practical lasers, usually a few closely spaced $\mathbf{k}$ modes become macroscopically excited, and the system can randomly jump from locking onto one mode to another due to the non-linear optical pumping which maintains the macroscopic mode occupation. It is the frequency of these jumps and the finite range of $\mathbf{k}$ values which limits the otherwise perfect optical coherence of a typical laser light source. See Loudon (1979) for a more detailed discussion of optical coherent states and their application to the laser.

3.4 Bosonic Quantum Fields

In this section we shall introduce **quantum field operators** for the case of boson particles. This will allow us to consider the quantum states of atomic BEC and superfluid ^4He . The BEC is a **weakly interacting** Bose system, while ^4He , as discussed in chapter 2, is a strongly interacting liquid of boson particles. At the same time we shall also see how the coherent state concept can also be applied to boson particles. This will enable us to define the macroscopic wave function $\psi(\mathbf{r})$ which we need in order to describe the condensate of particles. In this way we can generalize the simple intuitive approach of chapters 1 and 2 into something which is both systematic and rigorous.

Understanding BEC and superfluids obviously requires us to work with many-particle quantum states for very large numbers of particles. In elementary quantum mechanics we would write a Schrödinger equation for a wave function of N particles, in order to obtain an N body wave function

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N). \quad (3.42)$$

As discussed in Chapters 1 and 2, if we consider a system of N interacting Bose atoms then we could in principle write a wave function $\psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$ obeying the 3N-dimensional Schrödinger equation

$$\hat{H}\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) \quad (3.43)$$

where

$$\hat{H} = \sum_{i=1, N} \left(\frac{\hbar^2}{2m} \nabla_i^2 + V_1(\mathbf{r}) \right) + \frac{1}{2} \sum_{i,j=1, N} V(\mathbf{r}_i - \mathbf{r}_j). \quad (3.44)$$

Here $V_1(\mathbf{r})$ is the external potential, and $V(\mathbf{r})$ is the particle-particle interaction. In the case of ^4He this interaction could be taken as the Lennard-Jones potential of two helium atoms, Eq. ?? or Eq. ??, while for an atomic BEC we would use the delta function interaction of Eq. ?. The fact that the particles are bosons is expressed by the fact the the wave function must be symmetric under permutation of any two of the particle coordinates

$$\psi(\dots \mathbf{r}_i, \dots, \mathbf{r}_j \dots) = \psi(\dots \mathbf{r}_j, \dots, \mathbf{r}_i \dots) \quad (3.45)$$

representing an exchange of the identical particles at $\mathbf{r}_i$ and $\mathbf{r}_j$.

This approach is feasible for systems with a few particles, such as say the electrons in an atom, but it quickly becomes impractical in larger systems. A much more useful method is to adopt the methods of quantum field theory and introduce **field operators** which add or remove particles to the system. If we consider a single particle in a box we know that the wave functions are plane wave states

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{(V)^{1/2}} e^{i\mathbf{k} \cdot \mathbf{r}} \quad (3.46)$$

where V is the volume. We saw in chapter 1 that each of these single particle states can be occupied by 0, 1, 2, 3 or any other finite number of Bose particles. We denote these possible by the occupation number, $n_{\mathbf{k}}$. A general quantum many body state of the system will be a superposition of different N -body plane wave states. The complete basis of all possible states can be represented by the set of all possible occupation numbers of each plane wave. In exact analogy with the case of the laser we can creation and annihilation operators $a_{\mathbf{k}}^+$ and $a_{\mathbf{k}}$ which increase or decrease these occupation numbers. In order to satisfy the Bose symmetry condition on the wave function, Eq. 3.45 it turns out that it is necessary that

$$[a_{\mathbf{k}}, a_{\mathbf{k}}^+] = 1.$$

For two different plane wave states the occupation numbers are independent, and hence the creation operators must commute. Therefore the complete set of commutation relations are exactly as given in Eqs. 3.38-3.40. Similarly the occupation operator is given by the number operator,

$$\hat{n}_{\mathbf{k}} = a_{\mathbf{k}}^+ a_{\mathbf{k}} \quad (3.47)$$

These relations completely define the boson quantum field operators.

The set of many-particle states with all possible number operators, $\{n_{\mathbf{k}}\}$, is a complete set of wave functions. But, just as in the case of the harmonic oscillator, we can generate any $n_{\mathbf{k}}$ by successive actions of the operators $a_{\mathbf{k}}^+$, starting with a ground state $|0\rangle$. The interpretation is different now. $|0\rangle$ is the **vacuum state**, i.e. a state with no particles present. Successive actions of $a_{\mathbf{k}}^+$ add more particles to the system. Any many particle quantum state can be represented by a superposition of states generated in this way.

These field operators can also be cast into a real space form. We can define the quantum field operators $\hat{\psi}^+(\mathbf{r})$ and $\hat{\psi}(\mathbf{r})$ which create and annihilate particles at point $\mathbf{r}$. These can be defined by a Fourier transform of the $\mathbf{k}$ space operators,

$$\hat{\psi}(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} a_{\mathbf{k}} \quad (3.48)$$

$$\hat{\psi}^+(\mathbf{r}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} a_{\mathbf{k}}^+. \quad (3.49)$$

The function $e^{i\mathbf{k}\cdot\mathbf{r}}/\sqrt{V}$ here is obviously just a free particle plane wave in quantum state, $\mathbf{k}$. The inverse Fourier transforms are

$$\hat{a}_{\mathbf{k}} = \frac{1}{\sqrt{V}} \int e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{\psi}(\mathbf{r}) d^3r \quad (3.50)$$

$$\hat{a}_{\mathbf{k}}^+ = \frac{1}{\sqrt{V}} \int e^{i\mathbf{k}\cdot\mathbf{r}} \hat{\psi}^+(\mathbf{r}) d^3r \quad (3.51)$$

Using these definitions and the Bose commutation law, one can show (exercise 5.4) that these real-space field operators have the commutation laws,

$$[\hat{\psi}(\mathbf{r}), \hat{\psi}^+(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}') \quad (3.52)$$

$$[\hat{\psi}(\mathbf{r}), \hat{\psi}(\mathbf{r}')] = 0 \quad (3.53)$$

$$[\hat{\psi}^+(\mathbf{r}), \hat{\psi}^+(\mathbf{r}')] = 0. \quad (3.54)$$

We can also represent any operator in terms of its actions on quantum states described in terms of these operators. In particular the Hamiltonian, Eq. 3.44 becomes

$$\begin{aligned} \hat{H} &= \int \left(\hat{\psi}^+(\mathbf{r}) \left[\frac{\hbar^2}{2m} \nabla^2 + V_1(\mathbf{r}) \right] \hat{\psi}(\mathbf{r}) \right) d^3r \\ &\quad + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}) \hat{\psi}^+(\mathbf{r}') \hat{\psi}(\mathbf{r}') d^3r d^3r'. \end{aligned} \quad (3.55)$$

where the combination $\hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r})$ is obviously the density operator of particles at $\mathbf{r}$.

It turns out to be convenient to always work in “normal order”, in which all the creation operators are on the left and all the annihilation operators are on the right. Commuting two of the field operators above we obtain

$$\begin{aligned} \hat{H} &= \int \left(\hat{\psi}^+(\mathbf{r}) \left[\frac{\hbar^2}{2m} \nabla^2 + V_1(\mathbf{r}) \right] \hat{\psi}(\mathbf{r}) \right) d^3r \\ &\quad + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \hat{\psi}^+(\mathbf{r}) \hat{\psi}^+(\mathbf{r}') \hat{\psi}(\mathbf{r}) \hat{\psi}(\mathbf{r}') d^3r d^3r' \\ &\quad + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \hat{\psi}^+(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}') \hat{\psi}(\mathbf{r}') d^3r d^3r' \end{aligned} \quad (3.56)$$

The final term here arises from the commutator of $[\hat{\psi}(\mathbf{r}), \hat{\psi}^+(\mathbf{r}')]$, and reduces to

$$V(0) \int d^3r \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}) = V(0) \hat{N} \quad (3.57)$$

where

$$\hat{N} = \int d^3r \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}) \quad (3.58)$$

is obviously just the operator for the total number of particles in the system. The $\hat{N}V(0)$ term is a constant can be absorbed into the definition of the chemical potential, μ , and so we shall drop it from now on.

For a bulk fluid we can assume translational invariance, and ignore the external potential $V_1(\mathbf{r})$. Going back to $\mathbf{k}$ -space and we can use the Fourier transforms Eqs. 3.48 and 3.49 to represent the Hamiltonian in terms of $a_{\mathbf{k}}^+$ and $a_{\mathbf{k}}$. The kinetic energy term is

$$\begin{aligned} \hat{T} &= - \int \left(\hat{\psi}^\dagger(\mathbf{r}) \frac{\hbar^2}{2m} \nabla^2 \hat{\psi}(\mathbf{r}) \right) d^3r \\ &= \frac{1}{V} \sum_{\mathbf{k}\mathbf{k}'} \int \left(a_{\mathbf{k}'}^+ e^{-i\mathbf{k}' \cdot \mathbf{r}} \frac{\hbar^2 k'^2}{2m} a_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \right) d^3r \\ &= \sum_{\mathbf{k}} \frac{\hbar^2 k^2}{2m} a_{\mathbf{k}}^+ a_{\mathbf{k}}. \end{aligned} \quad (3.59)$$

The potential energy term is

$$\begin{aligned} \hat{V} &= \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}^\dagger(\mathbf{r}') \hat{\psi}(\mathbf{r}) \hat{\psi}(\mathbf{r}') d^3r d^3r' \\ &= \frac{1}{2V^2} \sum_{\mathbf{k}_1 \mathbf{k}_2 \mathbf{k}_3 \mathbf{k}_4} \int V(\mathbf{r} - \mathbf{r}') a_{\mathbf{k}_1}^+ a_{\mathbf{k}_2}^+ a_{\mathbf{k}_3} a_{\mathbf{k}_4} \\ &\quad e^{i(-\mathbf{k}_1 \cdot \mathbf{r} - \mathbf{k}_2 \cdot \mathbf{r}' + \mathbf{k}_3 \cdot \mathbf{r} + \mathbf{k}_4 \cdot \mathbf{r}')} d^3r d^3r' \\ &= \frac{1}{2V} \sum_{\mathbf{k}_1 \mathbf{k}_2 \mathbf{k}_3 \mathbf{k}_4} a_{\mathbf{k}_1}^+ a_{\mathbf{k}_2}^+ a_{\mathbf{k}_3} a_{\mathbf{k}_4} \delta_{\mathbf{k}_3 + \mathbf{k}_4, \mathbf{k}_1 + \mathbf{k}_2} \int V(\mathbf{r}) e^{i(\mathbf{k}_4 - \mathbf{k}_1) \cdot \mathbf{r}} d^3r. \end{aligned}$$

Introducing the Fourier transform of the interaction

$$V_{\mathbf{q}} = \frac{1}{V} \int V(r) e^{i\mathbf{q} \cdot \mathbf{r}} d^3r \quad (3.60)$$

and making the replacements $\mathbf{k}_1 \rightarrow \mathbf{k} + \mathbf{q}$, $\mathbf{k}_2 \rightarrow \mathbf{k}' - \mathbf{q}$, $\mathbf{k}_3 \rightarrow \mathbf{k}'$, and $\mathbf{k}_4 \rightarrow \mathbf{k}$ we can express the full Hamiltonian as

$$\hat{H} = \sum_{\mathbf{k}} \frac{\hbar^2 k^2}{2m} a_{\mathbf{k}}^+ a_{\mathbf{k}} + \frac{1}{2} \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}} V_{\mathbf{q}} a_{\mathbf{k}+\mathbf{q}}^+ a_{\mathbf{k}'-\mathbf{q}}^+ a_{\mathbf{k}'} a_{\mathbf{k}}. \quad (3.61)$$

We can interpret the interaction term simply as a process in which a pair of particles are scattered from initial states $\mathbf{k}$, $\mathbf{k}'$ to final states $\mathbf{k} + \mathbf{q}$, $\mathbf{k}' - \mathbf{q}$. The momentum transferred between the particles is $\mathbf{q}$, and the matrix element for the process is $V_{\mathbf{q}}$.

3.5 Off-Diagonal Long Ranged order

The field operators introduced in the previous section provide a general way to discuss quantum coherence in condensates and superfluids. Even though Eq. 3.61 is still too difficult to solve in general, we can still use it to explore the consequences of macroscopic quantum coherence in bosonic systems.

Firstly, let us revisit the idea of the macroscopic quantum state, as introduced for BEC and superfluid ^4He in chapters 1 and 2. Using the field operators we can redefine the one-particle density matrix as

$$\rho_1(\mathbf{r} - \mathbf{r}') \equiv \langle \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}') \rangle. \quad (3.62)$$

This definition is clearly more compact than the equivalent one given in Chapter 2. Using the Fourier transformations Eqs. 3.48 and 3.49 we find

$$\begin{aligned} \rho_1(\mathbf{r} - \mathbf{r}') &= \frac{1}{V} \sum_{\mathbf{k}, \mathbf{k}'} e^{i(\mathbf{k}' \cdot \mathbf{r}' - \mathbf{k} \cdot \mathbf{r})} \langle a_{\mathbf{k}}^+ a_{\mathbf{k}'} \rangle \\ &= \frac{1}{V} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} \langle a_{\mathbf{k}}^+ a_{\mathbf{k}} \rangle, \end{aligned} \quad (3.63)$$

which is just the Fourier transform of the momentum distribution

$$n_{\mathbf{k}} \equiv \langle a_{\mathbf{k}}^+ a_{\mathbf{k}} \rangle \quad (3.64)$$

exactly as found in chapter 2.

Now let us consider the consequences of these definitions in the case of a quantum coherent many-particle state. Just as in the case of the laser, we can define a coherent state

$$|\alpha_{\mathbf{k}_1}, \alpha_{\mathbf{k}_2}, \alpha_{\mathbf{k}_3} \dots\rangle$$

for any set of complex numbers $\alpha_{\mathbf{k}_i}$. Using the standard properties of coherent states we find

$$n_{\mathbf{k}} \equiv \langle |\alpha_{\mathbf{k}}|^2 \rangle \quad (3.65)$$

and hence

$$\rho_1(\mathbf{r} - \mathbf{r}') = \frac{1}{V} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} |\alpha_{\mathbf{k}}|^2. \quad (3.66)$$

Typically we will be interested in quantum states where only one of the $\mathbf{k}$ states is macroscopically occupied, (usually but not always $\mathbf{k} = 0$). So suppose that state $\mathbf{k}_0$ has occupation $N_0 = |\alpha_{\mathbf{k}_0}|^2$ where N_0 is a macroscopic number (a finite fraction of the total particle number N) and all the other $|\alpha_{\mathbf{k}_i}|^2$ are small. For such a state we will have the momentum distribution

$$n_{\mathbf{k}} = N_0 \delta_{\mathbf{k}, \mathbf{k}_0} + f(\mathbf{k}) \quad (3.67)$$

where $f(\mathbf{k})$ is a smooth function of $\mathbf{k}$. The corresponding density matrix is

$$\rho_1(\mathbf{r} - \mathbf{r}') = n_0 + \frac{2}{(2\pi)^3} \int d^3k e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} f(\mathbf{k}) \quad (3.68)$$

where $n_0 = N_0/V$.

These results are exactly the same as we found in chapter 2 by more elementary methods. The presence of the condensate is shown by the constant contribution, n_0 to the density matrix. If the function $f(\mathbf{k})$ is sufficiently smooth, then its Fourier transform will vanish for large $|\mathbf{r} - \mathbf{r}'|$ leaving just the constant contribution,

$$\langle \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}') \rangle \rightarrow n_0 \quad (3.69)$$

as $|\mathbf{r} - \mathbf{r}'| \rightarrow \infty$. This is what is meant by the term **off diagonal long ranged order** (ODLRO), introduced by Oliver Penrose.

Fig. 3.3 shows the physical interpretation of the ODLRO in superfluids. A particle can be annihilated at $\mathbf{r}$, and absorbed into the condensate, while a second particle is created at $\mathbf{r}'$ out of the condensate. This process has a quantum mechanical amplitude because of the quantum coherence of the condensate, even when the points $\mathbf{r}$ and $\mathbf{r}'$ are separated arbitrarily far apart. In contrast, for a normal liquid (even a normal quantum liquid) these processes would be incoherent except when $\mathbf{r}$ and $\mathbf{r}'$ are close together.

Using the coherent state concept, there is one more step which we can take. If the density matrix

$$\langle \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}') \rangle \quad (3.70)$$

is a constant, however far apart points $\mathbf{r}$ and $\mathbf{r}'$ are, then it seems plausible that we can treat the points as independent statistically. Then we can view the above as an average of a product of independent random variables and hence write it as a product of the two averages computed separately

$$\langle \hat{\psi}^+(\mathbf{r}) \hat{\psi}(\mathbf{r}') \rangle \rightarrow \langle \hat{\psi}^+(\mathbf{r}) \rangle \langle \hat{\psi}(\mathbf{r}') \rangle \quad (3.71)$$

for $|\mathbf{r} - \mathbf{r}'| \rightarrow \infty$.

If we were to work in the standard fixed particle number, N , many-body formulation of quantum mechanics then averages such as $\langle \hat{\psi}^+(\mathbf{r}) \rangle$ would be automatically zero, since acting on any N particle state $\hat{\psi}^+(\mathbf{r})|N\rangle$ is an $N + 1$ particle state, and is necessarily orthogonal to $\langle N|$. But if we are working in the basis of coherent states, then there is no such problem. The coherent states have definite phase, not definite N , and this type of expectation value is perfectly well allowed.

Therefore we can say that there is an **order parameter** or **macroscopic wave function**, defined by

$$\psi_0(\mathbf{r}) = \langle \hat{\psi}(\mathbf{r}) \rangle. \quad (3.72)$$

In terms of this function we see that⁴

$$\rho_1(\mathbf{r} - \mathbf{r}') = \psi_0^*(\mathbf{r}) \psi_0(\mathbf{r}') \quad (3.73)$$

⁴The creation operator $\hat{\psi}^+(\mathbf{r})$ is just the Hermitian conjugate of $\hat{\psi}(\mathbf{r})$ and so $\psi_0^*(\mathbf{r}) = \langle \hat{\psi}^+(\mathbf{r}) \rangle$.

for $|\mathbf{r} - \mathbf{r}'| \rightarrow \infty$. In a translationally invariant system (the condensation occurs in the $\mathbf{k} = 0$ state), we must therefore have

$$\psi_0(\mathbf{r}) = \sqrt{n_0} e^{i\theta} \quad (3.74)$$

where θ is an arbitrary constant phase angle.

Of course this phase θ is nothing more than the XY model phase angle introduced in chapter 2, Fig. ???. But now we can see that its true meaning is that we have a coherent quantum state in which the $\mathbf{k} = 0$ state has a macroscopic occupation.

Since we have not yet made any connection to the Hamiltonian, Eq. 3.61 it is impossible to prove from these arguments that such a coherent quantum state will be stable. But at least we can see how to construct coherent many-particle wave functions in which a definite order parameter phase θ is possible. In the case of the ideal Bose condensate discussed in chapter 1 one can still work in the fixed particle number representation, and so there is no advantage to explicitly introduce a coherent state formalism. But as soon as there are any interactions, however weak, the coherent state approach becomes advantageous. In the next section we shall consider the weakly interacting Bose gas, in which the advantages of the coherent state approach can be seen explicitly.

3.6 The Weakly Interacting Bose Gas

The theory of the **weakly interacting Bose gas** was originally developed by Bogoliubov in in late 1940's. It was developed as a theory of superfluid helium, although as we have seen, for ^4He the interatomic interactions are very strong. In this case the theory has some qualitative features which agree with experimental properties of ^4He , most notably the linear phonon like quasiparticle excitation spectrum, $\epsilon_{\mathbf{k}} = ck$, at small wave vectors in fig. ???. But it fails to reproduce other important features, such as the roton minimum in the spectrum. On the other hand, the theory is believed to be close to exact for the case of atomic BEC, since the conditions under which it is derived are close to the experimental ones.

First of all we shall assume that we are at zero temperature, or close to zero, so that the system is close to its ground state. We assume that the system is in a coherent many-particle state, characterized by a macroscopic wave function $\psi_0(\mathbf{r})$, as in Eq. 3.72. Suppose that the many-particle quantum state $|\psi\rangle$, is an ideal coherent state at zero temperature. Then it is an eigenstate of the annihilation operator,

$$\hat{\psi}(\mathbf{r})|\psi\rangle = \psi_0(\mathbf{r})|\psi\rangle. \quad (3.75)$$

We can view this as a trial many-particle wave function, and we will vary the parameter $\psi_0(\mathbf{r})$ so as to variationally minimize the total energy. The

variational energy is found by taking the expectation value of the Hamiltonian,

$$\begin{aligned}\hat{H} &= \int \hat{\psi}^+(\mathbf{r}) \left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) \right) \hat{\psi}(\mathbf{r}) d^3r \\ &\quad + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \hat{\psi}^+(\mathbf{r}) \hat{\psi}^+(\mathbf{r}') \hat{\psi}(\mathbf{r}) \hat{\psi}(\mathbf{r}') d^3r d^3r'. \quad (3.76)\end{aligned}$$

Here the single particle potential $V_1(\mathbf{r})$ is the effective external potential of the atom trap. In the case of bulk superfluids this is obviously zero.

Using the definition of the coherent state $|\psi\rangle$ from Eq. 3.75 we can find immediately that the variational energy is

$$\begin{aligned}E_0 &= \langle \psi | \hat{H} | \psi \rangle \\ &= \int \psi_0^*(\mathbf{r}) \left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) \right) \psi_0(\mathbf{r}) d^3r \\ &\quad + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') \psi_0^*(\mathbf{r}) \psi_0^*(\mathbf{r}') \psi_0(\mathbf{r}) \psi_0(\mathbf{r}') d^3r d^3r'. \quad (3.77)\end{aligned}$$

We can find the minimum by functional differentiation, exactly as for the Ginzburg-Landau equation. Setting

$$\frac{\partial E_0}{\partial \psi_0^*(\mathbf{r})} = 0$$

yields,

$$\left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) - \mu \right) \psi_0(\mathbf{r}) + \int V(\mathbf{r} - \mathbf{r}') \psi_0(\mathbf{r}) \psi_0^*(\mathbf{r}') \psi_0(\mathbf{r}') d^3r' = 0. \quad (3.78)$$

The parameter μ is a Lagrange multiplier, necessary to maintain a constant normalization of the macroscopic wave function

$$N_0 = \int |\psi_0(\mathbf{r})|^2 d^3r. \quad (3.79)$$

Clearly Eq. 3.78 is of the form of an effective Schrödinger equation

$$\left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) + V_{eff}(\mathbf{r}) - \mu \right) \psi_0(\mathbf{r}) = 0 \quad (3.80)$$

where μ is the chemical potential, and the effective potential is

$$V_{eff}(\mathbf{r}) = \int V(\mathbf{r} - \mathbf{r}') |\psi_0^*(\mathbf{r}')|^2 d^3r'$$

This Schrödinger equation is exactly the Gross-Pitaevski equation again, which we derived by a different method in chapter 2.

To examine the accuracy of this ground state, and to examine the low energy excited states, we need to consider possible quantum states which are close to our trial ground state $|\psi\rangle$, but which do not deviate from it too much. We need to consider many-particle states which do not exactly obey the coherent state condition, Eq. 3.75, but for which it is nearly obeyed. Bogoliubov introduced an elegant method to achieve this. He assumed that the field operators can be expressed approximately as their constant coherent state value, plus a small deviation,

$$\hat{\psi}(\mathbf{r}) = \psi_0(\mathbf{r}) + \delta\hat{\psi}(\mathbf{r}). \quad (3.81)$$

From the commutation relations for the field operators, it is easy to see that

$$[\delta\hat{\psi}(\mathbf{r}), \delta\hat{\psi}^\dagger(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'), \quad (3.82)$$

and so the deviation operators $\delta\hat{\psi}(\mathbf{r})$ and $\delta\hat{\psi}^\dagger(\mathbf{r})$ are also bosonic quantum fields.⁵ We can then rewrite the hamiltonian in terms of $\psi_0(\mathbf{r})$ and $\delta\hat{\psi}(\mathbf{r})$. We can group terms according to whether $\delta\hat{\psi}(\mathbf{r})$ occurs never, once, twice, three times or four,

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2 + \dots \quad (3.83)$$

and we assume that it is valid to simply ignore terms higher than second order.

In this expansion, the first term, $\hat{H}_0$ is simply the original coherent state energy, Eq. 3.77 which we can minimize using the Gross-Pitaevskii equations. Furthermore, if we have variationally minimized the energy there will be no corrections to the energy to linear order in the deviation operators $\delta\hat{\psi}(\mathbf{r})$. the first significant correction term is quadratic in the deviation operators. Several terms contribute, but the net result is that

$$\begin{aligned} \hat{H}_2 = & \int \delta\hat{\psi}^\dagger(\mathbf{r}) \left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) \right) \delta\hat{\psi}(\mathbf{r}) d^3r \\ & + \frac{1}{2} \int V(\mathbf{r} - \mathbf{r}') d^3r d^3r' \left(\delta\hat{\psi}^\dagger(\mathbf{r}) \delta\hat{\psi}^\dagger(\mathbf{r}') \psi_0(\mathbf{r}) \psi_0(\mathbf{r}') \right. \\ & + 2\delta\hat{\psi}^\dagger(\mathbf{r}) \psi_0^*(\mathbf{r}') \delta\hat{\psi}(\mathbf{r}) \psi_0(\mathbf{r}') + 2\psi_0^*(\mathbf{r}) \delta\hat{\psi}^\dagger(\mathbf{r}') \delta\hat{\psi}(\mathbf{r}) \psi_0(\mathbf{r}') \\ & \left. + \psi_0^*(\mathbf{r}) \psi_0^*(\mathbf{r}') \delta\hat{\psi}(\mathbf{r}) \delta\hat{\psi}(\mathbf{r}') \right). \end{aligned} \quad (3.84)$$

One can visualize the meaning of these various terms in terms of the simple diagrams shown in Fig. 3.4. There are four distinct terms. The first, shown in panel (a), corresponds to the creation of two particles, one at $\mathbf{r}$ and $\mathbf{r}'$ under the action of the potential $V(\mathbf{r} - \mathbf{r}')$. Of course they are

⁵A different way to see this is to imagine that we simply translate the origin of the coherent state complex plane, Fig. 3.2. Shifting the origin from $\alpha = 0$ to $\alpha = \psi_0(\mathbf{r})$ we can then describe states which are near to $|\psi\rangle$ in terms of the coherent states α which are in the vicinity of the point $\psi_0(\mathbf{r})$.

not really created, but scattered out of the condensate. The second term corresponds to the scattering of an existing quasiparticle by interaction with particles in the condensate. It has a extra factor of 2 since we can find an identical diagram with $\mathbf{r}$ and $\mathbf{r}'$ interchanged. The third diagram (c) is also a scattering of an existing quasiparticle, but now the quasiparticle at $\mathbf{r}$ is absorbed into the condensate, while at the same time a second quasiparticle appears at $\mathbf{r}'$. Again $\mathbf{r}$ and $\mathbf{r}'$ can be interchanged, leading to an extra factor of 2. The final diagram, (d), shows two quasiparticles being absorbed into the condensate.

To keep the algebra manageable let us specialize to the case (relevant for the atomic BEC) of a pure contact interaction,

$$V(\mathbf{r} - \mathbf{r}') = g\delta(\mathbf{r} - \mathbf{r}'). \quad (3.85)$$

For convenience we will also assume that $\psi_0(\mathbf{r})$ is real, and equal to $\psi_0(\mathbf{r}) = \sqrt{n_0(\mathbf{r})}$, where $n_0(\mathbf{r})$ the spatially varying condensate density in the atom trap. The quadratic Hamiltonian $\hat{H}_2$ simplifies to

$$\begin{aligned} \hat{H}_2 = & \int \delta\hat{\psi}^+(\mathbf{r}) \left(-\frac{\hbar^2 \nabla^2}{2m} + V_1(\mathbf{r}) - \mu \right) \delta\hat{\psi}(\mathbf{r}) d^3r \\ & + \frac{g}{2} \int n_0(\mathbf{r}) \left(\delta\hat{\psi}^+(\mathbf{r})\delta\hat{\psi}^+(\mathbf{r}) + 4\delta\hat{\psi}^+(\mathbf{r})\delta\hat{\psi}(\mathbf{r}) + \delta\hat{\psi}(\mathbf{r})\delta\hat{\psi}(\mathbf{r}) \right) d^3r. \end{aligned} \quad (3.86)$$

This Hamiltonian is a quadratic form in the operators, and it turns out that all such quadratic Hamiltonians can be diagonalized exactly. The procedure makes use of the **Bogoliubov** transformation to eliminate the “anomalous” terms $\delta\hat{\psi}(\mathbf{r})\delta\hat{\psi}(\mathbf{r})$ and $\delta\hat{\psi}^+(\mathbf{r})\delta\hat{\psi}^+(\mathbf{r})$. Define a new pair of operators by

$$\hat{\varphi}(\mathbf{r}) = u(\mathbf{r})\delta\hat{\psi}(\mathbf{r}) + v(\mathbf{r})\delta\hat{\psi}^+(\mathbf{r}) \quad (3.87)$$

$$\hat{\varphi}^+(\mathbf{r}) = u^*(\mathbf{r})\delta\hat{\psi}^+(\mathbf{r}) + v^*(\mathbf{r})\delta\hat{\psi}(\mathbf{r}). \quad (3.88)$$

These are again bosonic quantum field operators if

$$[\hat{\varphi}(\mathbf{r}), \hat{\varphi}^+(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'), \quad (3.89)$$

which is true provided that the functions $u(\mathbf{r})$ and $v(\mathbf{r})$ are chosen to obey

$$|u(\mathbf{r})|^2 - |v(\mathbf{r})|^2 = 1. \quad (3.90)$$

The general solution becomes quite complicated, so let us specialize to the case of a uniform Bose liquid, without the atom trap potential $V(\mathbf{r})$. Assuming that the macroscopic wave function is also just a constant, $\psi_0 = \sqrt{n_0}$, and going to $\mathbf{k}$ -space Eq. 3.86 becomes

$$\hat{H}_2 = \sum_{\mathbf{k}} \left(\left(\frac{\hbar^2 k^2}{2m} - \mu \right) a_{\mathbf{k}}^+ a_{\mathbf{k}} + \frac{n_0 g}{2} (a_{\mathbf{k}}^+ a_{-\mathbf{k}}^+ + 4a_{\mathbf{k}}^+ a_{\mathbf{k}} + a_{-\mathbf{k}} a_{\mathbf{k}}) \right). \quad (3.91)$$

The Bogoliubov transformation in k -space gives the new operators

$$b_{\mathbf{k}} = u_{\mathbf{k}} a_{\mathbf{k}} + v_{\mathbf{k}} a_{-\mathbf{k}}^+ \quad (3.92)$$

$$b_{\mathbf{k}}^+ = u_{\mathbf{k}}^* a_{\mathbf{k}} + v_{\mathbf{k}}^* a_{-\mathbf{k}} \quad (3.93)$$

where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ obey

$$|u_{\mathbf{k}}|^2 - |v_{\mathbf{k}}|^2 = 1. \quad (3.94)$$

The idea is to rewrite the Hamiltonian in terms of these new operators and then to vary the parameters $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ to make it diagonal. In particular it is necessary to make the coefficients of the anomalous terms, $b_{\mathbf{k}}^+ b_{-\mathbf{k}}^+$ and $b_{-\mathbf{k}} b_{\mathbf{k}}$ equal to zero. Since the calculation is lengthy we shall just quote the results. It turns out that when the anomalous terms are eliminated, the new quasiparticle excitations created by the $b_{\mathbf{k}}^+$ operators have the energy spectrum

$$E_{\mathbf{k}} = \left(\frac{\hbar^2 k^2}{2m} \right)^{1/2} \left(\frac{\hbar^2 k^2}{2m} + 2n_0 g \right)^{1/2}. \quad (3.95)$$

For small $|\mathbf{k}|$ (less than $\sim \sqrt{4n_0 g m}/\hbar$) the spectrum is linear

$$E_{\mathbf{k}} \sim ck \quad (3.96)$$

where the “phonon” velocity is

$$c = \left(\frac{\hbar^2 n_0 g}{m} \right)^{1/2}. \quad (3.97)$$

This Bogoliubov quasiparticle spectrum is sketched in Fig. 3.5. The spectrum is linear at small $\mathbf{k}$, and joins smoothly onto the independent particle energy $\hbar^2 k^2/2m$ for large $\mathbf{k}$. The success of this Bogoliubov theory is that it explains the linear dispersion in the excitation spectrum near to $\mathbf{k} = 0$, as we saw in the case of superfluid ^4He , Fig. ???. In chapter 2 we have also seen that a linear spectrum is necessary to prevent quasiparticle scattering from the container walls, and hence to maintain a dissipationless superflow. Therefore we can conclude that even a very weakly interacting Bose gas is a superfluid, even when the ideal non-interacting Bose gas is not. The critical velocity for the superfluid will be less than c , and so by Eq. 3.97 the critical velocity will approach zero in the limits of weak interaction $g \rightarrow 0$, or low density $n_0 \rightarrow 0$.

Of course there are still huge differences between the Bogoliubov quasiparticle spectrum in Fig. 3.5 and the experimental quasiparticle spectrum of Fig. ??. Most importantly, there is no roton minimum. The Bogoliubov spectrum is linear at small $\mathbf{k}$ and becomes equal to the free particle spectrum $\hbar^2 k^2/2m$ for large $\mathbf{k}$, but it has no minimum at intermediate $\mathbf{k}$. The roton minimum is therefore an effect emerging only in a strongly interacting Bose liquid. This difference has another effect, namely that the Bogoliubov spectrum $E_{\mathbf{k}}$ has a slight upwards curvature at small $\mathbf{k}$, while the true spectrum has a slight downward curvature. This curvature means that a quasiparticle of momentum $\hbar\mathbf{k}$ and energy $E_{\mathbf{k}}$ has a non-zero cross section

to decay into three quasiparticles of lower energy and momentum. Therefore the Bogoliubov quasiparticle is not an exact eigenstate, but only a long lived resonance.⁶

3.7 Coherence and ODLRO in Superconductors

The above ideas were originally introduced to explain the strongly interacting Bose superfluid ^4He . But they also apply to superconductors. However in this case the argument must be modified in order to take account of the fact that the electrons in a superconductor are fermions. Although it is perfectly possible to define coherent states for fermions, they are not immediately useful for superconductivity. This is because a single fermion state can only ever be occupied by either 0 or 1 fermions, due to the exclusion principle. Therefore it is not possible to have a macroscopic number of fermions in a single plane wave state.

It was Robert Schrieffer who first managed to write down a coherent many-particle wave function for fermions. His colleagues Bardeen and Cooper had already realized that electrons bind into pairs in a superconductor. There was even an earlier theory by Schafroth Blatt and Butler in which superconductivity was seen as a Bose condensate of electron pairs.⁷ But Bardeen Cooper and Schrieffer (BCS) knew that the pairs of electrons in superconductors could not be simply treated as bosons. The problem was to write down a valid many-body wave function for the electrons in which each electron participate in the pairing. The brilliant solution Schrieffer discovered was effectively another type of coherent state, similar to those we have already seen. But the key point is to have a coherent state in which a macroscopic number of **pairs** are all in the same state.

Nevertheless even in the BCS theory we have a form of ODLRO and an order parameter, as we shall discover in this section. Their appearance is a quite general phenomenon. Here we concentrate on the most general statements about the ODLRO, and leave the actual details of the BCS theory, and its specific predictions until the next chapter.

⁶It makes perfect sense here to borrow the concepts of elementary particle physics, and to talk about one particle decaying into a set of others. Just as in particle physics we can call such ‘particles’ resonances. In this context the Bogoliubov quasiparticles are just the ‘elementary particles’ of the Bose gas, and the background condensate is the analog of the vacuum.

⁷Unfortunately this theory was not able to make quantitative predictions for the superconductors which were known at that time, and it was generally discarded in favour of the Bardeen Cooper Schrieffer theory which was very much more successful in making quantitative predictions. The electron pairs in the Bardeen Cooper and Schrieffer theory **are not bosons**. In general a pair of fermions is not equivalent to a boson. In the BCS case the bound electron pairs are very large, and so different pairs strongly overlap each other. In this limit it is not possible to describe the pair as a boson, and so the BCS theory is not normally expressed in terms of Bose condensation. However in recent years theories based upon the ideas of the Schafroth, Blatt and Butler theory have been somewhat revived as a possible models of high T_c superconductors.

First we must define the correct quantum field operators to describe the conduction electrons in a solid. In single-particle quantum mechanics of solids we know that the wave functions are Bloch waves

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r}). \quad (3.98)$$

Here the crystal wave vector $\mathbf{k}$ must lie within the first Brillouin zone. For simplicity of notation we shall assume that only one of the energy bands, n , is relevant (i.e. the one at the Fermi surface) and so we will drop the index n from now on.

A particular Bloch state of a given spin σ , $\psi_{\mathbf{k}\sigma}(\mathbf{r})$, can either be empty or occupied by an electron. A quantum state of N particles can be specified by saying for each individual state whether it is occupied or not. This is **the occupation number representation** of quantum mechanics. We can introduce operators which change these occupation numbers. Suppose we label the state as $|0\rangle$ or $|1\rangle$ if the given Bloch state $\psi_{\mathbf{k}\sigma}(\mathbf{r})$. Then we can define operators which change the occupation numbers

$$\begin{aligned} c^+|0\rangle &= |1\rangle \\ c|1\rangle &= |0\rangle. \end{aligned}$$

These are obviously similar to the similar Bose field operators, or the Harmonic oscillator ladder operators, $\hat{a}^+$ and $\hat{a}$. The only difference is that the state of a harmonic oscillator can have $n = 0, 1, 2, \dots$, while for fermions the occupation number is only 0 or 1. The operators c^+ and c are again called the 'creation' and 'annihilation' operators. Since electrons are neither being created or annihilated in solids these names may seem a bit misleading, however the operators work the same way in particle physics where particles are indeed being created or annihilated back into the vacuum (eg. electron-positron pairs annihilating each other). In a solid one can imagine adding electrons to the solid (e.g. with an external current source into the surface), or removing them from the solid (eg. in photo-emission).

These operators have a couple of important properties. Firstly, just as for bosons, the combination c^+c measures the occupation number $|n\rangle$ since

$$\begin{aligned} c^+c|0\rangle &= 0 \\ c^+c|1\rangle &= |1\rangle \end{aligned}$$

i.e. $c^+c|n\rangle = n|n\rangle$. Secondly the exclusion principle implies that we cannot occupy a state with more than one fermion, and hence $c^+c^+|n\rangle = 0$ and $cc|n\rangle = 0$. This fermion nature of the particles means that

$$\{c, c^+\} \equiv cc^+ + c^+c = 1 \quad (3.99)$$

where $\{A, B\} = AB + BA$ is the **anti-commutator** of operators A and B . The antisymmetry of the many-particle wave function for fermions means

that the operators for different Bloch states or spin states **anti-commute**, and so in general we have the anti-commutation relations,

$$\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}^{\dagger}\} = \delta_{\mathbf{k}\sigma, \mathbf{k}'\sigma'} \quad (3.100)$$

$$\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}\} = 0 \quad (3.101)$$

$$\{c_{\mathbf{k}\sigma}^{\dagger}, c_{\mathbf{k}'\sigma'}^{\dagger}\} = 0 \quad (3.102)$$

where $\sigma = \pm 1$ denotes the two different spin states. Naturally, just as for bosons, we can also represent these operators in real space, by a Fourier transformation, obtaining

$$\{\hat{\psi}_{\sigma}(\mathbf{r}), \hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}')\} = \delta(\mathbf{r} - \mathbf{r}')\delta_{\sigma\sigma'} \quad (3.103)$$

$$\{\hat{\psi}_{\sigma}(\mathbf{r}), \hat{\psi}_{\sigma'}(\mathbf{r}')\} = 0 \quad (3.104)$$

$$\{\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}), \hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}')\} = 0. \quad (3.105)$$

In some sense this is also like a Bose condensate of the Cooper pairs. However because the fermion antisymmetry is fully taken account of the wave function leads to quite different predictions than one might just expect with a Bose condensation picture. The BCS predictions turned out to be extremely accurate numerically, despite it being a simplified mean-field model wave function. Its accuracy lies in the large size of the Cooper pairs, with their characteristic size ξ (the coherence length) being much larger than the typical inter-electron spacings in a solid r_s , where $N/V = n = 4\pi r_s^3/3$.

Now the mathematical challenge which BCS needed to solve was to write down a many-particle wave function in which every electron near the Fermi surface participates in the pairing. They knew that a single pair of electrons would bind into a spin singlet state with two body wave function

$$\Psi(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2) = \varphi(\mathbf{r}_1 - \mathbf{r}_2) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (3.106)$$

They first wrote down a many-particle wave function in which every particle is paired,

$$\begin{aligned} \Psi(\mathbf{r}_1\sigma_1, \dots, \mathbf{r}_N\sigma_N) &= \frac{1}{\sqrt{N!}} \sum_P (-1)^P \Psi(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2) \Psi(\mathbf{r}_3\sigma_3, \mathbf{r}_4\sigma_4) \dots \\ &\dots \Psi(\mathbf{r}_{N-1}\sigma_{N-1}, \mathbf{r}_N\sigma_N). \end{aligned} \quad (3.107)$$

Here the sum over P denotes the sum over all the $N!$ permutations of the N particle labels $\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2$ etc. The sign $(-1)^P$ is positive for an even permutation and -1 for odd permutations. This alternating sign is necessary so that the many-body wave function has the correct fermion antisymmetry

$$\Psi(\dots, \mathbf{r}_i\sigma_i, \dots, \mathbf{r}_j\sigma_j, \dots) = -\Psi(\dots, \mathbf{r}_j\sigma_j, \dots, \mathbf{r}_i\sigma_i, \dots). \quad (3.108)$$

But this fixed N many-body quantum mechanics is unwieldy. By fixing N we cannot have a definite overall phase, unlike in a coherent state representation. But if we use coherent states then it is possible to describe a condensate with a definite phase, and the key step achieved by Schrieffer was to find a way to write down a coherent state of fermion pairs. First we need to write down operators which create or annihilate pairs of electrons. Defining,

$$\hat{\varphi}^+(\mathbf{R}) \equiv \int d^3r \int \varphi(\mathbf{r}) \hat{\psi}_{\uparrow}^+(\mathbf{R} + \mathbf{r}/2) \hat{\psi}_{\downarrow}^+(\mathbf{R} - \mathbf{r}/2). \quad (3.109)$$

we can see that acting with this operator on a quantum state with N electrons gives a new quantum state with $N + 2$ electrons. It creates a spin singlet electron pair, where the electrons are separated by $\mathbf{r}$ and with centre of mass at $\mathbf{R}$.⁸

Naively one might regard such a fermion pair as a boson. But this is not correct. If we try to evaluate the commutator we find that

$$[\hat{\varphi}(\mathbf{R}), \hat{\varphi}^+(\mathbf{R}')] \neq \delta(\mathbf{R} - \mathbf{R}') \quad (3.110)$$

$$[\hat{\varphi}(\mathbf{R}), \hat{\varphi}(\mathbf{R}')] \neq 0 \quad (3.111)$$

$$[\hat{\varphi}^+(\mathbf{R}), \hat{\varphi}^+(\mathbf{R}')] \neq 0. \quad (3.112)$$

The operators only commute when $\mathbf{R}$ and $\mathbf{R}'$ are far apart, corresponding to non-overlapping pairs. For this reason we cannot simply make a Bose condensate out of these pairs.

But even though these are not true boson operators, we can still define the analogue of ODLRO corresponding to Bose condensation. Now it is a state of ODLRO of **Cooper pairs**. We can define a new density matrix by,

$$\rho_1(\mathbf{R} - \mathbf{R}') = \langle \hat{\varphi}^+(\mathbf{R}), \hat{\varphi}(\mathbf{R}') \rangle. \quad (3.113)$$

This is a one particle density matrix for pairs, and so it is related to the two particle density matrix for the electrons

$$\rho_2(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2, \mathbf{r}_3\sigma_3, \mathbf{r}_4\sigma_4) = \langle \hat{\psi}_{\sigma_1}^+(\mathbf{r}_1) \hat{\psi}_{\sigma_2}^+(\mathbf{r}_2) \hat{\psi}_{\sigma_3}(\mathbf{r}_3) \hat{\psi}_{\sigma_4}(\mathbf{r}_4) \rangle. \quad (3.114)$$

Inserting the definition of the pair operator gives the pair density matrix in terms of the electron one

$$\rho_1(\mathbf{R} - \mathbf{R}') = \int \varphi(\mathbf{r}) \varphi(\mathbf{r}') \rho_2(\mathbf{R} + \frac{\mathbf{r}}{2} \uparrow, \mathbf{R} - \frac{\mathbf{r}}{2} \downarrow, \mathbf{R}' - \frac{\mathbf{r}'}{2} \downarrow, \mathbf{R}' + \frac{\mathbf{r}'}{2} \uparrow) d^3r d^3r'. \quad (3.115)$$

The pair wave function $\varphi(\mathbf{r})$ is a quantum mechanical bound state, and so it will become zero for large $|\mathbf{r}|$. If the scale of its range is defined by a length,

⁸Note that we assume here that the wave function for the bound electron pair obeys $\varphi(\mathbf{r}) = \varphi(-\mathbf{r})$. This is simply because a spin singlet bound state wave function must be even under exchange of particle coordinates, because the spin single state is odd under exchange.

ξ_0 , (which turns out to be the BCS coherence length of the superconductor), then the the main contributions to the pair density matrix comes from the parts of electron density matrix where $\mathbf{r}_1$ and $\mathbf{r}_2$ are separated by less than ξ_0 , and similarly $\mathbf{r}_3$ and $\mathbf{r}_4$ are separated by less than ξ_0 . But the pair $\mathbf{r}_1, \mathbf{r}_2$ and the pair $\mathbf{r}_3, \mathbf{r}_4$ can be separated by any arbitrarily large distance.

Now we can have ODLRO in the pair density matrix, provided that

$$\rho_1(\mathbf{R} - \mathbf{R}') \rightarrow \text{const.} \quad (3.116)$$

as $|\mathbf{R} - \mathbf{R}'| \rightarrow \infty$. The BCS theory does therefore correspond to a similar macroscopic quantum coherence to the ordinary theory of ODLRO in super-fluids. However is is an ODLRO of Cooper pairs, not single electrons.

In terms of the electron density matrix $\rho_2(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2, \mathbf{r}_3\sigma_3, \mathbf{r}_4\sigma_4)$ this ODLRO corresponds to the density matrix approaching a constant value in a limit where the two coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ are close to each other, and $\mathbf{r}_3$ and $\mathbf{r}_4$ are close, but these two pairs are separated by an arbitrarily large distance. Fig. 3.6 illustrates this concept.

Following the same approach to ODLRO as in the weakly interacting Bose gas, we can make the assumption that very distant points $\mathbf{R}$ and $\mathbf{R}'$ should behave independently. Therefore we should be able to write

$$\rho_1(\mathbf{R} - \mathbf{R}') \sim \langle \hat{\varphi}^+(\mathbf{R}) \rangle \langle \hat{\varphi}(\mathbf{R}') \rangle \quad (3.117)$$

for $|\mathbf{R} - \mathbf{R}'| \rightarrow \infty$. We can therefore also define a **macroscopic wave function** by

$$\psi(\mathbf{R}) = \langle \hat{\varphi}(\mathbf{R}) \rangle. \quad (3.118)$$

Effectively this is the **Ginzburg-Landau order parameter** for the superconductor.

Of course, we have still not actually shown how to construct a many-electron quantum state which would allow this type of Cooper pair ODLRO. We leave that until the next chapter. Nevertheless we can view the above discussion as stating the requirements for the sort of quantum state which could exhibit superconductivity. It was possibly the most significant achievement of the BCS theory that it was possible to explicitly construct such a non-trivial many-body quantum state. In fact we can see immediately from Eq. 3.118 that we must have a coherent state with a definite quantum mechanical phase θ , and conversely that we should not work with fixed particle number N . At the time of the original publication of the BCS theory in 1957 this aspect was one of the most controversial of the whole theory. The explicit appearance of the phase θ also caused concern, since it appeared to violate the principle of gauge invariance. It was only the near perfect agreement between the predictions of the BCS theory and experiments, as well as clarification of the gauge invariance issue by Anderson and others which led to the final acceptance of the BCS theory.

3.8 The Josephson Effect

The Josephson effect is a direct physical test of the quantum coherence implied by superconducting ODLRO. Soon after the BCS theory was published Brian Josephson, then a young PhD student at Cambridge, considered the effect of electrons tunnelling between two different superconductors.⁹

Consider two superconductors, separated by a thin insulating layer, as shown in Fig. 3.7. If each superconductor has a the macroscopic wave function as defined in, 3.118, then we can assign definite values to the wave functions on either side of the tunnel barrier, say ψ_L and ψ_R for the left and right hand side respectively,

$$\begin{aligned}\psi_L(\mathbf{R}_L) &= \langle L | \hat{\varphi}(\mathbf{R}_L) | L \rangle \\ \psi_R(\mathbf{R}_R) &= \langle R | \hat{\varphi}(\mathbf{R}_R) | R \rangle\end{aligned}$$

where $\mathbf{R}_L$ and $\mathbf{R}_R$ are points on either side of the tunnel barrier, and $|L\rangle$ and $|R\rangle$ are the many-particle quantum states of the superconductors on the left and right side respectively. Josephson assumed that electron tunnelling takes place for electrons crossing the barrier. In terms of electron field operators we can write

$$\hat{H} = \sum_{\sigma} \int T(\mathbf{r}_L, \mathbf{r}_R) (\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}_L) \hat{\psi}_{\sigma}(\mathbf{r}_R) + \hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}_R) \hat{\psi}_{\sigma}(\mathbf{r}_L)) d^3r_L d^3r_R \quad (3.119)$$

as the operator which tunnels electrons of spin σ from points $\mathbf{r}_L$ and $\mathbf{r}_R$ on either side of the barrier. Using the BCS many-body wave functions on either side of the junction and second order perturbation theory in the tunnelling Hamiltonian $\hat{H}$, Josephson was able to find the remarkable result that a current flows in the junction, given by

$$I = I_c \sin(\theta_1 - \theta_2) \quad (3.120)$$

where θ_1 and θ_2 are the phases of the macroscopic wave functions on either side of the junction.

The details of how Josephson found this result will not be important here. But it is worth noting very roughly how this might come about. If we consider effects to second order in the tunnelling Hamiltonian $\hat{H}$ we can see that $\hat{H}^2$ contains many terms, but includes some four fermion terms of the following form,

$$\hat{H}^2 \sim T^2 (\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}_L) \hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}'_L) \hat{\psi}_{\sigma}(\mathbf{r}_R) \hat{\psi}_{\sigma'}(\mathbf{r}'_R))$$

⁹Brian Josephson received the Nobel prize in 1973 for this discovery, possibly one of the few Nobel prizes to have arisen from a PhD project. After Josephson's first prediction of the effect in 1962, established experts in the superconductivity field at first objected to the theory, believing that the effect was either not present or would be too weak to observe. But Josephson's PhD adviser Anderson encouraged Rowell to look for the effect experimentally. Anderson and Rowell together announced the first observation of the Josephson effect in 1963.

$$+\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}_R)\hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}'_R)\hat{\psi}_{\sigma}(\mathbf{r}_L)\hat{\psi}_{\sigma'}(\mathbf{r}'_L)+\dots). \quad (3.121)$$

The net effect of the first of these terms is to transfer a pair of electrons from the right hand superconductor to the left. Conversely the second term transfers a pair from left to right. Because the many-body states on both sides of the junction are coherent pair states, these operators will have non-zero expectation values consistent with the ODLRO

$$\begin{aligned} \langle L|\hat{\psi}_{\sigma}^{\dagger}(\mathbf{r}_L)\hat{\psi}_{\sigma'}^{\dagger}(\mathbf{r}'_L)|L\rangle &\neq 0 \\ \langle R|\hat{\psi}_{\sigma}(\mathbf{r}_R)\hat{\psi}_{\sigma'}(\mathbf{r}'_R)|R\rangle &\neq 0. \end{aligned} \quad (3.122)$$

$$(3.123)$$

In fact we expect the first of these expectation values proportional to $e^{-i\theta_1}$ and the second to $e^{i\theta_2}$. The overall quantum mechanical amplitude for tunnelling a pair from right to left is thus has a phase $e^{i(\theta_2-\theta_1)}$. The reverse process, tunnelling from left to right has the opposite phase. When they are added together the net current is proportional to $\sin(\theta_1 - \theta_2)$ as given by Eq. 3.120.

Eq. 3.120 shows that the current flows in response to the phase difference $\theta_1 - \theta_2$. Therefore it is in some sense a direct proof of the existence of such coherent state phases in superconductors. The proportionality constant I_c is the maximum Josephson current that can flow, and is called the **critical current** of the junction.

For currents I below I_c the Josephson current is perfectly dissipationless, i.e. it is a supercurrent. But if current is driven to a higher value, $I > I_c$, a finite voltage drop V develops across the junction. Therefore the typical $I - V$ characteristic of the junction is as shown in Fig. 3.8.

In the case that $I > I_c$ Josephson found a second surprising consequence of this tunnel current. The finite voltage difference V between the superconductors, means that the macroscopic wave functions become time dependent. Using a version of the Heisenberg equation of motion for the left and right hand side macroscopic wave functions,

$$\begin{aligned} i\hbar \frac{\partial \psi_L(t)}{\partial t} &= -2eV_L \psi_L(t) \\ i\hbar \frac{\partial \psi_R(t)}{\partial t} &= -2eV_R \psi_R(t), \end{aligned} \quad (3.124)$$

Josephson was able to show that the finite voltage drop $V = V_L - V_R$ leads to a steadily increasing phase difference,

$$\Delta\theta(t) = \Delta\theta(0) + \frac{2eV}{\hbar}t \quad (3.125)$$

and hence the Josephson current,

$$I = I_c \sin\left(\Delta\theta(0) + \frac{2eV}{\hbar}t\right) \quad (3.126)$$

oscillates at a frequency

$$\nu = \frac{2eV}{\hbar}. \quad (3.127)$$

This surprising effect is called the a.c. Josephson effect (in contrast to the d.c. Josephson effect for $I < I_c$, $V = 0$).

The experimental observation of the a.c. Josephson effect not only confirmed the theory and the validity of the BCS macroscopic quantum coherent state, but also provided another direct empirical confirmation that the relevant particle charge is $2e$ and not e . Thus it again confirmed the Cooper pairing hypothesis. Even more surprising is that the Josephson frequency appears to be **exactly** given by Eg. 3.127. In fact, it is so accurate that the Josephson effect has been incorporated into the standard set of measurements used to define the SI unit system. By measuring the frequency (which can be measured with accuracies of one part in 10^{12} or better) and the voltage V one can obtain the ratio of fundamental constants $e/\hbar$ with high precision. Alternatively one can use the given values of $e/\hbar$ and the Josephson effect to define a reliable and portable voltage standard, V .

The Josephson effect is also at the heart of many different practical applications of superconductivity. One of the simplest devices to make is a SQUID ring; where SQUID stands for Superconducting QUantum Interference Device. This is simply a small (or large) superconducting ring in which there are two weak links. Each half of the ring is then connected to external leads, as shown in Fig. 3.9. By “weak link” one can mean either a tunnel barrier, such as Fig. 3.7 (an SIS junction), or a thin normal metallic spacer (an SNS junction). The current through each junction depends on the phase difference across it and so,

$$I = I_{c1} \sin(\Delta\theta_1) + I_{c2} \sin(\Delta\theta_2) \quad (3.128)$$

is the sum of the currents of each Josephson junction. The phases differences $\Delta\theta_1$, $\Delta\theta_2$ correspond to the macroscopic wave function phases differences at the points to the left and right of each junction in Fig. 3.9.

In the junctions are perfectly balanced, so $I_{c1} = I_{c2}$ and a small external current I ($< I_c$) is applied to the SQUID, then we would expect an equal steady state current to flow in both halves of the ring, and a constant phase difference equal to $\Delta\theta = \sin^{-1}(I/2I_c)$ will develop across both junctions. But this is only true if there is no magnetic flux through the ring. Using the principle of gauge invariance we find that a flux Φ implies that the phase differences are no longer equal.

$$\begin{aligned} \Phi &= \int \mathbf{B} \cdot d\mathbf{S} \\ &= \oint \mathbf{A} \cdot d\mathbf{r} \\ &= \frac{2e}{\hbar} \oint (\nabla\theta) \cdot d\mathbf{r} \end{aligned}$$

$$\begin{aligned}
&= \frac{2e}{\hbar} (\Delta\theta_1 - \Delta\theta_2) \\
&= 2\pi\Phi_0 (\Delta\theta_1 - \Delta\theta_2). \tag{3.129}
\end{aligned}$$

Therefore the magnetic flux through the ring leads to a difference between the two phases. For a balanced SQUID ring system we can assume that

$$\begin{aligned}
\Delta\theta_1 &= \Delta\theta + \frac{\pi\Phi}{\Phi_0} \\
\Delta\theta_2 &= \Delta\theta - \frac{\pi\Phi}{\Phi_0}. \tag{3.130}
\end{aligned}$$

therefore the total current in the SQUID is

$$\begin{aligned}
I &= I_c \sin(\Delta\theta_1) + I_c \sin(\Delta\theta_2) \\
&= I_c \sin\left(\Delta\theta + \frac{\pi\Phi}{\Phi_0}\right) + I_c \sin\left(\Delta\theta - \frac{\pi\Phi}{\Phi_0}\right) \\
&= 2I_c \sin(\Delta\theta) \cos\left(\frac{\pi\Phi}{\Phi_0}\right). \tag{3.131}
\end{aligned}$$

The critical current is therefore modulated by a factor depending on the net flux through the ring,¹⁰

$$I_c(\Phi) = I_0 \left| \cos\left(\frac{\pi\Phi}{\Phi_0}\right) \right|. \tag{3.132}$$

This modulation of the observed SQUID ring critical current is shown in Fig. 3.10. This current is essentially an ideal Fraunhofer interference pattern, exactly analogous to the interference pattern one observes in optics with Young's two slit experiment. Here the two Josephson junctions are playing the role of the two slits, and the interference is between the supercurrents passing through the two halves of the ring. The supercurrents acquire different phases due to the magnetic field. One can say that this effect is also analogous to the Ahronov Bohm effect (Feynman 1964), in which a single electron passes on either side of a of a solenoid of magnetic flux.

The SQUID device provides a simple, but highly accurate, system for measuring magnetic flux. Since the flux quantum Φ_0 is only about $2 \times 10^{-15} \text{Wb}$ in the SI unit system, and one can make SQUID devices approaching 1cm^2 in area, it is in principle possible to measure magnetic fields to an accuracy below $B \sim 10^{-10} \text{T}$. In particular it is easy to measure **changes** in field to this accuracy by simply counting the number of minima in the SQUID critical current.

¹⁰The SQUID ring critical current will always have the same sign as the driving current and hence the modulus signs appearing in Eq. 3.132.

3.9 Macroscopic Quantum Coherence

To what extent does the Josephson effect or the SQUID represent true evidence quantum coherence? Even though it is very cold, operating at a temperature below T_c , say at one or two degrees Kelvin, a SQUID ring is hardly isolated from its environment. In fact one can make perfectly good SQUIDS using high temperature superconductors. These operate at temperatures of over 100K. SQUID rings are usually fabricated on some sort of insulating substrate, and are usually subject to normal external electromagnetic noise in the laboratory, unless well shielded.

Given this relatively noisy thermal environment, the SQUID shows a remarkable insensitivity to these effects. This is fundamentally because the macroscopic wave function $\psi(\mathbf{r})$ which we have defined above, and its phase θ , is not a true wave function in the sense of elementary quantum mechanics. In particular it does not obey the fundamental **principle of superposition**, and one cannot apply the usual **quantum theory of measurement** or Copenhagen interpretation to it. The macroscopic wave function behaves much more like a thermodynamic variable, such as the magnetization in a ferromagnet, than a pure wave function, even though it has a phase and obeys local gauge invariance.

But since the early 1980's there have been attempts to observe true quantum superpositions in superconductors (Leggett 1980). I.e. can one construct a "Schrödinger cat" like quantum state? For example is a state such as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\psi_1\rangle + |\psi_2\rangle) \quad (3.133)$$

meaningful in a SQUID ring? If $|\psi_1\rangle$ and $|\psi_2\rangle$ are two pure quantum states, then the general principle of linearity of quantum mechanics, implies that any superposition such as $|\psi\rangle$ must also be a valid quantum state. Only by measuring some observable can one "collapse" the wave function and find out whether the system was in $|\psi_1\rangle$ or $|\psi_2\rangle$. for small system, such as single atoms or photons, such superpositions are a standard part of quantum mechanics. But in his famous 1935 paper Schrödinger showed that this fundamental principle leads to paradoxes with our everyday understanding of the world when we apply it to macroscopic systems such as the famous cat in a box!

Even since Schrödinger's paper it has not been clear where to put the dividing line between the "macroscopic" world (governed by classical physics and without superposition) and the microscopic (governed by quantum mechanics). The ideas of **decoherence** provide one possible route by which quantum systems can acquire classical behaviour. Interactions with the environment lead to entanglement between the quantum states of the system and those of the environment, and (in some of the most modern approaches) "quantum information" is lost.

In this context one can say that a large SQUID ring, say 1cm (or 100m!) in diameter will be subject to decoherence from its environment, and hence

will be effectively in the classical realm. But if one makes the ring smaller, or operates at lower temperatures, is there a regime where true quantum superpositions occur? In fact the answer to this question is yes! Indeed strong evidence for quantum superposition states has now been seen in three different systems.

The first system where quantum superposition states were observed was in Bose-Einstein condensates, in 1996 (Ketterle 2002). Since these exist in a very low temperature state (a micro Kelvin or less) and are isolated from most external thermal noise sources (since they are trapped in vacuum) one could expect a high degree of quantum coherence to occur. Indeed it has proved possible to “split” a single condensate into two halves, in a similar way to which a beam splitter separates photons. When the two halves of the condensate are subsequently brought back together again, then one observes an interference pattern. The experiment is effectively the exact analogue of the Young’s slit interference experiment with light.

The second type of experiment which showed true quantum interference was done using a superconducting island or “Cooper pair box”. This consists of a small island of superconductor (Al was used) of order $0.1\mu\text{m}$ on a side, as shown in Fig. 3.12. Operating at temperatures of a few mK, well below the superconducting T_c , the quantum states of the box can be characterized entirely by the number of Cooper pairs present. For example the box can have a state $|N\rangle$ of N Cooper pairs, or a state $|N + 1\rangle$ etc. The energies of these different states can be manipulated through external voltage gates, since they are states of different total electronic charge on the box. The analogue of the Schrodinger cat state for this system is to place it in a superposition, such as

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|N\rangle + |N + 1\rangle). \quad (3.134)$$

Nakamura Pashkin and Tsai (1999) were able to demonstrate the presence of just such superpositions in their device. By connecting the Cooper pair box to a second superconductor, via a Josephson junction, they effectively allowed quantum mechanical transitions between these two states, as Cooper pairs tunnel onto or off the island. By pulsing external voltage gates connected to the system they observed beautiful interference fringes associated with the superposition states, as shown in Fig. 3.13. The figure shows the final charge on the box (i.e. $N + 1$ or N) as a function of the voltage pulse amplitude and duration. The results observed oscillations agree excellently with the theoretical predictions based on the existence of macroscopic quantum superposition states.

The third systems in which macroscopic quantum coherence has been demonstrated are small superconducting rings, as in Fig. 2.6. As we saw in chapter 3, a superconducting ring has a set of different ground states corresponding to different winding numbers of the order parameter around the ring. These can again be represented by abstract many-particle states

such as $|n\rangle$ and $|n+1\rangle$. In a large ring there will be no way for the system to tunnel from one of these states to another, but if the ring is made small enough (below $1\mu\text{m}$ in diameter) such transitions become possible. Two experiments in the year 2000 observed direct evidence for quantum mechanical coherence in such systems (van der Wal *et al.* 2000, (Friedman *et al.* 2000)). (Earlier reports unconfirmed reports of the effect were also made). In any case these results are fascinating, since they represent a coherent quantum tunnelling of a system containing 10^{10} or more electrons. Fig. 3.14 shows an electron microscope image of a small superconducting circuit, approximately $2\mu\text{m}$ across, superimposed on an image of the quantum (Rabi) oscillations observed in this circuit (Chiorescu *et al.* 2003). These oscillations demonstrate the existence of quantum superposition states in which two the system is simultaneously in two macroscopically different quantum states!

Do these quantum superposition states have any practical uses? In recent years there has been a huge growth in the field of **quantum information** and **quantum computation**. The idea is that “information” as used and manipulated in computer bits is actually always a physical quantity, e.g. the charges on the capacitors in a RAM computer memory. Therefore it is subject to the laws of physics. But conventional computer bits are essentially based on classical physics. For example a computer bit can be measured without disturbing its state. But if we imagine eventually making the physical computer bits smaller and smaller with each new generation of computer, then eventually we will have to use devices which are so small that quantum mechanics applies, not classical physics. For such a quantum bit, or **qubit**, information is carried by its full quantum state, not just by a classical 0 or 1. Surprisingly it even turns out the computers based on manipulation of these qubits would be far more efficient than classical computers for certain types of algorithms. But whether this goal can be ever achieved depends on finding suitable physical systems in which to realize the qubit. While there are many possibilities under active investigation, superconducting devices or BEC have several possible advantages for these types of problem. At the very least, the experiments described above demonstrating macroscopic quantum superposition states show that BEC or superconducting devices are at least one viable option for a physical qubit. See Mahklin Schön and Shnirman (2001) and Annett Gyorffy and Spiller (2002) for more discussion of possible superconducting qubit devices.

3.10 Summary

In this chapter we have explored the implications of **quantum coherent states** in the theory of Bose and Fermi systems. We have seen how Bose coherent states provide an effective way to understand the laser and the weakly interacting Bose gas. The key point being that the coherent state representation allows one to discuss quantum states with definite phase

θ , rather than with definite particle number N . Using the coherent state approach the ideas of an effective macroscopic quantum wave function and off diagonal long ranged order (ODLRO) also become quite natural.

For fermion systems the coherent state approach is also quite natural, provided that one deals with coherent states of Cooper pairs, not single electrons. We have not yet seen how to explicitly construct such a coherent state (see the next chapter!), but we have already been able to see how ODLRO and the Ginzburg-Landau order parameter both arise naturally from this formalism. The Josephson effect, and its applications to SQUID devices can also be understood qualitatively even without the full description of the BCS wave function.

Finally we have seen that both BEC and superconductors do indeed exhibit **macroscopic quantum coherence**. But in the case of superconductors this is only evident when the devices are small enough and cold enough to avoid the effects of decoherence. Although the usual Josephson effect and the SQUID interference patterns are both interference effects, they do not in themselves show the existence of quantum superposition states such as the Schrödinger cat.

3.11 Further Reading

See Loudon (1979) for a more detailed discussion of optical coherent states and their application to the laser. A more advanced and general review of all applications of coherent states is given by Klauder and Skagerstam (1985).

The theory of the weakly interacting Bose gas is discussed in detail in Pines (1961), a book which also includes a reprint of the original paper by Bogoliubov (1947). More mathematically advanced approaches using many-body Green's function techniques are given by Fetter and Walecka (1971), and the Abrikosov, Gorkov and Dzyaloshinski (1963).

P.W Anderson made many key contributions to the development of the ideas of ODLRO and macroscopic coherence in superconductors. His book *Basic Notions in Condensed Matter Physics*, (Anderson 1984), includes several reprints of key papers in the discovery of ODLRO in superconductors, the Josephson effect and related topics. Tinkham (1996) also has a very detailed discussion about the Josephson effect and SQUID devices.

The problems and paradoxes posed by macroscopic quantum coherence are discussed by Leggett (1980), with a recent up date in Leggett (2002). The possibilities of making superconducting qubit devices for quantum computation are discussed in Mahklin Schön and Shnirman (2001) and Annett Gyorffy and Spiller (2002).

3.12 Exercises

(5.1) (a) Using the definitions of the ladder operators given in Eq. 3.12 show that

$$[\hat{a}, \hat{a}^+] = 1$$

and

$$\hat{H} = \hbar\omega_c \left(\hat{a}^+ \hat{a} + \frac{1}{2} \right).$$

(b) Show that

$$[\hat{H}, \hat{a}^+] = \hbar\omega_c \hat{a}^+.$$

Hence show that if $\psi_n(x)$ is an eigenstate of the Hamiltonian with energy E_n , then ψ_{n+1} (defined by Eq. 3.2) is also an eigenstate with energy

$$E_{n+1} = E_n + \hbar\omega_c.$$

(c) Assuming that $\psi_n(x)$ is normalized, show that ψ_{n+1} as defined by Eq. 3.2 is also a correctly normalized quantum state.

(5.2) Using the fundamental defining equation of the coherent state Eq. 3.13, show that two coherent states $|\alpha\rangle$ and $|\beta\rangle$ have the overlap,

$$\langle\alpha|\beta\rangle = e^{-|\alpha|^2/2} e^{-|\beta|^2/2} e^{\alpha^* \beta},$$

and hence derive Eq. 3.33.

(5.3) Show that for a coherent state $|\alpha\rangle$

$$\langle\alpha|((\hat{a}^+)^p \hat{a}^q|\alpha\rangle = (\alpha^*)^p \alpha^q$$

for any positive integers p and q .

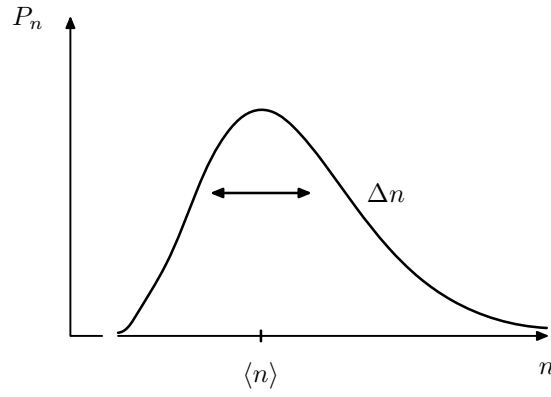


Fig. 3.1 The probability of the a coherent state containing quantum number N is a Poisson distribution. The width Δn is of order $\sqrt{\langle n \rangle}$, and so $\Delta n/\sqrt{\langle n \rangle} \rightarrow 0$ for large $\langle n \rangle$. Therefore Δn becomes negligible for coherent states with macroscopic numbers of particles.

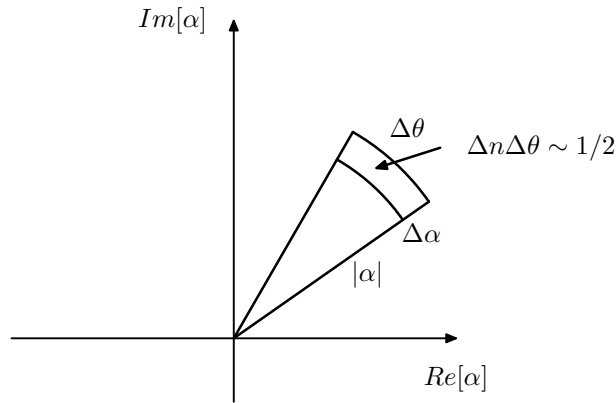


Fig. 3.2 Complex plane of coherent states, $|\alpha\rangle$, with $\alpha = |\alpha|e^{i\theta}$. The area $\Delta\theta\Delta n$ contains approximately one independent orthogonal quantum state.

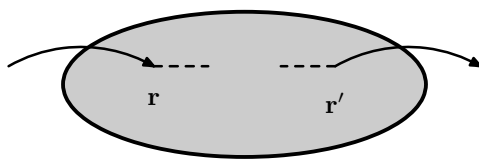


Fig. 3.3 Schematic illustrating the interpretation of the one particle density matrix $\rho_1(\mathbf{r} - \mathbf{r}')$. A particle is inserted into the condensate at $\mathbf{r}$, and a particle is removed from it at $\mathbf{r}'$. In a condensate, this process has a coherent quantum amplitude and phase however great the separation between $\mathbf{r}$ and $\mathbf{r}'$.

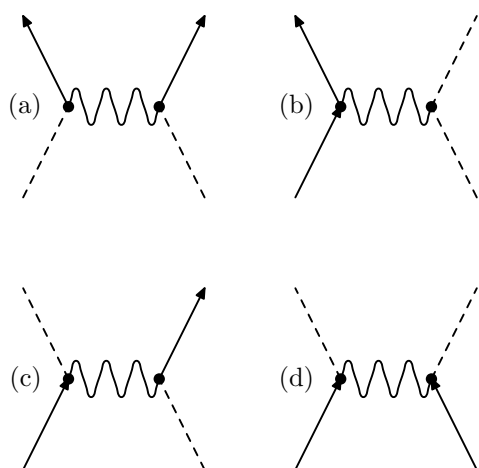


Fig. 3.4 Four types of interactions between quasiparticles and a Bose condensate. The quasiparticles are denoted by the solid lines, the condensate particles by the dashed line, and the interaction $V(\mathbf{r})$ by the wavy line. (a) Two particles are excited out of the condensate. (b) An existing quasiparticle interacts with the condensate. (c) An existing quasiparticle is absorbed into the condensate, while simultaneously a second quasiparticle becomes excited out of it. (d) Two quasiparticles are absorbed into the condensate.

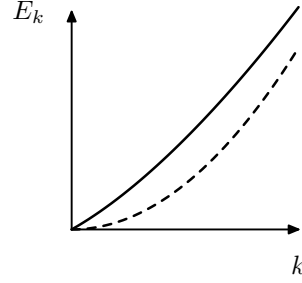


Fig. 3.5 The quasiparticle spectrum of a weakly interacting Bose gas, as found by Bogoliubov. The spectrum is linear at small $\mathbf{k}$, and approaches the free particle energy $\hbar^2 k^2 / 2m$ for large $\mathbf{k}$. Unlike the case of superfluid ^4He , Fig. ?? there is no roton minimum, and there is a slight upward curvature near to $\mathbf{k} = 0$.

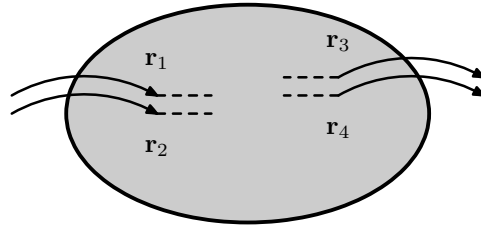


Fig. 3.6 The two body density matrix for electrons in a metal, $\rho_2(\mathbf{r}_1\sigma_1, \mathbf{r}_2\sigma_2, \mathbf{r}_3\sigma_3, \mathbf{r}_4\sigma_4)$. Off diagonal long ranged order (ODLRO) for electron pairs appears when this has a finite value however far apart the pair $\mathbf{r}_1$ and $\mathbf{r}_2$, is from $\mathbf{r}_3$ and $\mathbf{r}_4$. In contrast, the points within each pair must be no more than a coherence length, ξ_0 apart.

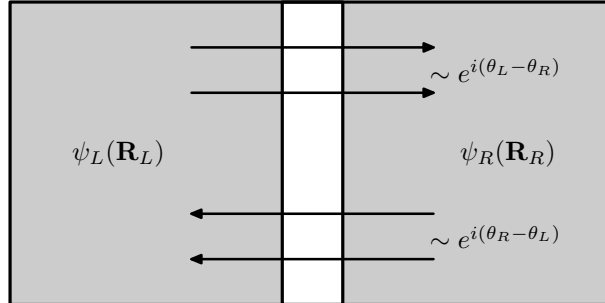


Fig. 3.7 A schematic Josephson tunnel junction between two superconductors. To second order in the tunnelling Hamiltonian there are two possible processes for a Cooper pair to tunnel from one side to another. If θ_L and θ_R denote the order parameter phases on either the left and right hand sides of the junction, then the amplitudes for processes depend on $e^{-i(\theta_L - \theta_R)}$ and $e^{-i(\theta_R - \theta_L)}$. When these are added together the net tunnel current is proportional to $\sin(\theta_L - \theta_R)$.

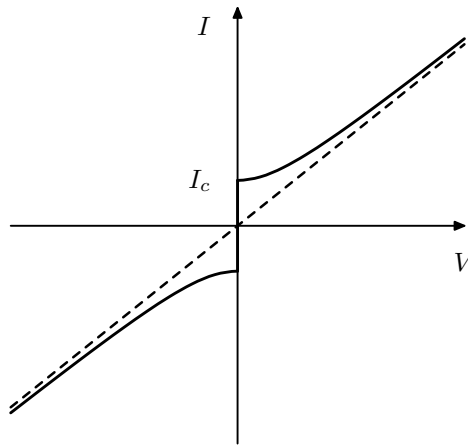


Fig. 3.8 I-V characteristic of a Josephson junction. There is no voltage drop, $V = 0$, provided that the junction current is less than the critical current I_c . Above this value a finite voltage drop occurs. This approaches the normal state Ohm's law result $I = V/R$ for large currents. The a.c. Josephson effect occurs in the $V \neq 0$ regime above I_c .

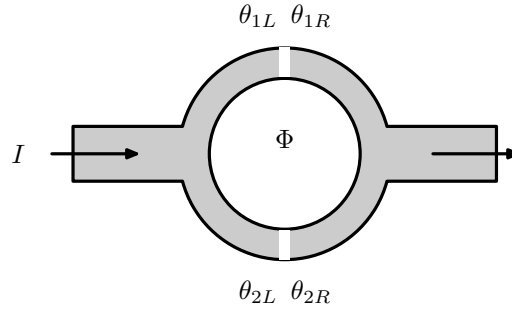


Fig. 3.9 Schematic geometry of a SQUID ring. The two Josephson junctions have currents governed by the phase differences $\Delta\theta_1 = \theta_{1L} - \theta_{1R}$ and $\Delta\theta_2 = \theta_{2L} - \theta_{2R}$. The total critical current of the whole device is modulated by the total magnetic flux through the ring Φ .

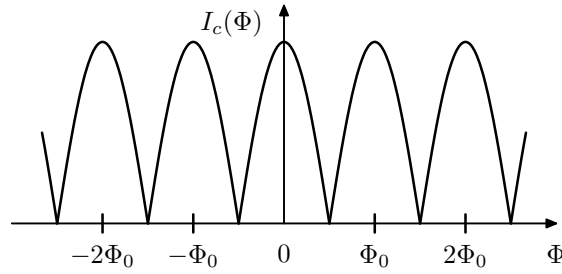


Fig. 3.10 Modulation of critical current in a SQUID ring. This is effectively equivalent to the Fraunhofer interference pattern of a two slit optical interference pattern. Effectively there is interference between the two currents flowing through the opposite sides of the SQUID ring in Fig. 3.9.

Fig. 3.11 Macroscopic quantum coherence demonstrated in a Bose-Einstein condensate. A condensate is split into two halves, which then interfere with each other, analogous to an optical beam splitter experiment. The interference fringes are clearly visible as horizontal bands of light and dark absorption, corresponding to a spatially modulated condensate density. Reproduced from Ketterle (2002) with permission.

Fig. 3.12 An electron micrograph image of the Cooper pair box device used to demonstrate macroscopic quantum coherence in superconductors. The Cooper pair box, is connected to its environment via Josephson coupling to the charge reservoir, as indicated, and to a probe device, which is used to measure the number of Cooper pairs, N on the box. The device is manipulated through the two electrical gates indicated, one providing a d.c. bias, and the second delivering pico-second pulses which switch the device from one quantum state to another. Reproduced from Nakamura, Pashkina and Tsai (1999) with permission.

Fig. 3.13 Quantum oscillations of charge observed in the Cooper pair box of fig. 3.12. The measured device current, I , is proportional to the Cooper pair number N on the box, and so the oscillations demonstrate quantum superpositions of states $|N\rangle$ and $|N + 1\rangle$. The oscillations depend on the amplitude and duration of the gate pulses, corresponding to the two axes shown in the diagram. Reproduced from Nakamura, Pashkin and Tsai (1999) with permission.

Fig. 3.14 A small superconducting SQUID type circuit, approximately $2\mu\text{m}$ across. This device shows coherent oscillations which are direct consequences of quantum superposition states. In the two states a current either circulates clockwise or anti-clockwise, as illustrated by the arrows drawn. Because the ring is macroscopic (containing or order 10^{10} superconducting electrons) this demonstrates the existence of “Schrödinger cat” quantum superposition states of macroscopically different states. reproduced from Chiorescu *et al.* (2003) with permission.